

Proof. **Preliminary cases.**

If $n = 1$ then both sides of (C) equal $a_1 b_1$, so the inequality holds with equality.

If $a_1 = a_2 = \cdots = a_n = a$ (or similarly for the b -sequence), then

$$\sum_{i=1}^n a_i b_i = a \sum_{i=1}^n b_i, \quad n \sum_{i=1}^n a_i b_i = \left(\sum_{i=1}^n a_i \right) \left(\sum_{i=1}^n b_i \right),$$

so (C) is an equality. Hence we may assume $n \geq 2$ and that neither sequence is constant.

Reformulation in symmetric form.

Define the arithmetic means

$$\bar{a} := \frac{1}{n} \sum_{i=1}^n a_i, \quad \bar{b} := \frac{1}{n} \sum_{i=1}^n b_i,$$

and consider

$$\Sigma := \sum_{i=1}^n (a_i - \bar{a})(b_i - \bar{b}). \quad (2.1)$$

Expanding and collecting terms gives

$$\begin{aligned} \Sigma &= \sum_{i=1}^n a_i b_i - \bar{a} \sum_{i=1}^n b_i - \bar{b} \sum_{i=1}^n a_i + n \bar{a} \bar{b} \\ &= \sum_{i=1}^n a_i b_i - \frac{1}{n} \left(\sum_{i=1}^n a_i \right) \left(\sum_{i=1}^n b_i \right). \end{aligned} \quad (2.2)$$

Thus (C) is equivalent to

$$\Sigma \geq 0. \quad (2.3)$$

Rearrangement inequality argument.

Rearrangement Inequality (standard form). If $x_1 \leq x_2 \leq \cdots \leq x_n$ and $y_1 \leq y_2 \leq \cdots \leq y_n$, then for every permutation $\sigma \in S_n$,

$$\sum_{i=1}^n x_i y_i \geq \sum_{i=1}^n x_i y_{\sigma(i)}. \quad (3.1)$$

The identity permutation yields the maximal possible sum. Applying (3.1) with $x_i = a_i$ and $y_i = b_i$ and averaging over all permutations we obtain

$$\sum_{i=1}^n a_i b_i \geq \frac{1}{n!} \sum_{\sigma \in S_n} \sum_{i=1}^n a_i b_{\sigma(i)}. \quad (3.2)$$

For a fixed pair (i, j) the term $a_i b_j$ appears exactly $(n-1)!$ times in the double sum, because we may fix $\sigma(i) = j$ and permute the remaining $n-1$ indices arbitrarily. Hence

$$\sum_{\sigma \in S_n} \sum_{i=1}^n a_i b_{\sigma(i)} = (n-1)! \sum_{i=1}^n \sum_{j=1}^n a_i b_j = (n-1)! \left(\sum_{i=1}^n a_i \right) \left(\sum_{j=1}^n b_j \right).$$

Dividing by $n!$ gives

$$\frac{1}{n!} \sum_{\sigma \in S_n} \sum_{i=1}^n a_i b_{\sigma(i)} = \frac{1}{n} \left(\sum_{i=1}^n a_i \right) \left(\sum_{j=1}^n b_j \right). \quad (3.3)$$

Insert (3.3) into (3.2) and multiply by n :

$$n \sum_{i=1}^n a_i b_i \geq \left(\sum_{i=1}^n a_i \right) \left(\sum_{j=1}^n b_j \right),$$

which is precisely inequality (C). By (2.2) this also shows $\Sigma \geq 0$.

Equality conditions.

The Rearrangement Inequality is strict unless the two sequences are similarly ordered **and** at least one of them is constant. Consequently, equality in (C) occurs iff one of the sequences (a_i) or (b_i) is constant. When both sequences contain at least two distinct elements the inequality is strict.

Conclusion.

We have treated the trivial cases, rewritten Chebyshev's sum inequality in a symmetric covariance form, and proved its non-negativity by averaging over all permutations (a direct consequence of the Rearrangement Inequality). This establishes (C) for all $n \geq 1$ and characterises the equality cases. $\square$ $\square$